

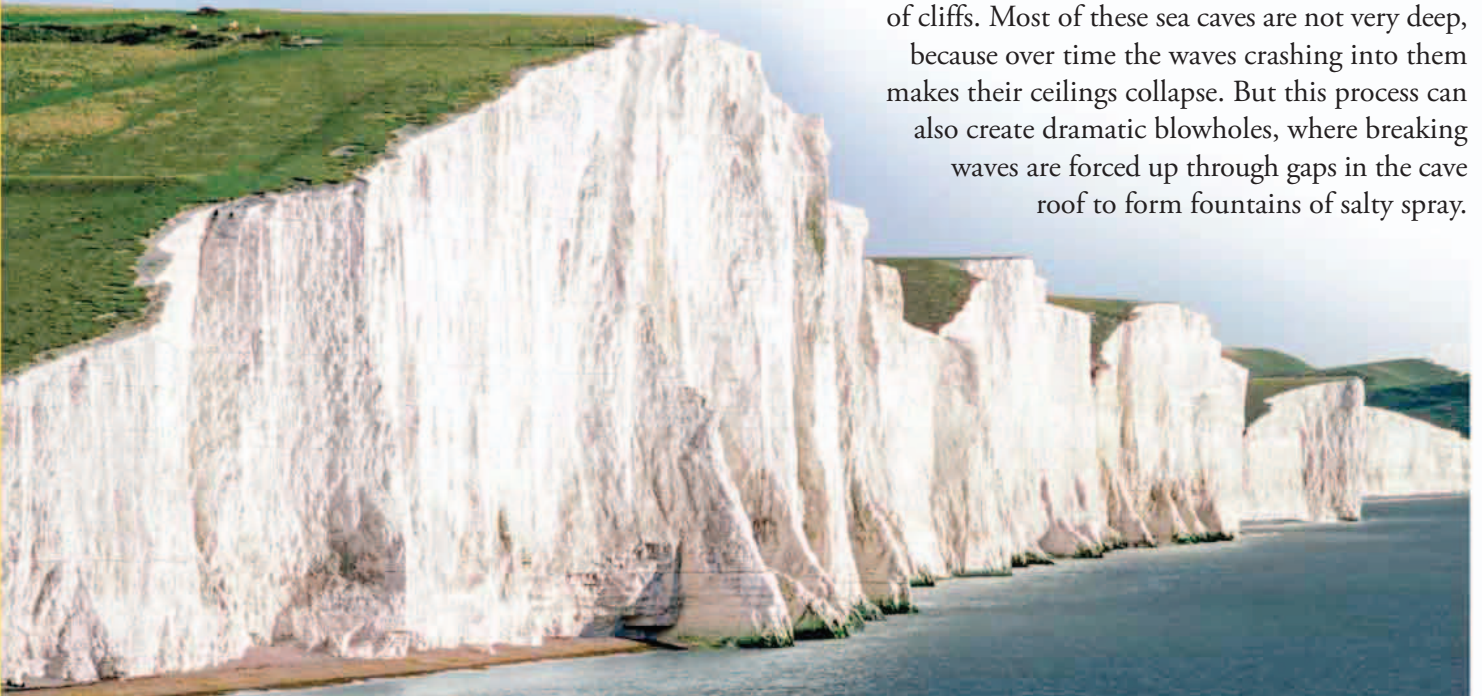


# Cliffs and caves

As ocean waves break on rocky shores, they shatter and crumble the rock, and sweep the debris away to other shores. This relentless process creates a range of spectacular coastal features, including sheer cliffs, dark caves, soaring arches, isolated islands, and tall stacks. But as fast as these features are created, others are destroyed by the same forces.

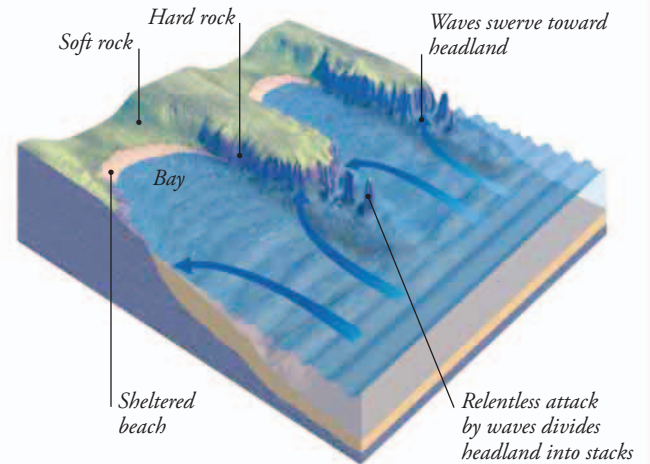
## SHEER DROP

Where high land meets the sea, the rock is cut back at sea level. This leaves the rock above without any support so it collapses under its own weight, creating a cliff. The shape of the cliff varies according to the type of rock, but the most dramatic sheer cliffs usually form in softer rock such as chalk. These chalk cliffs in southern England are known as the Seven Sisters.



## BAYS AND HEADLANDS

Where the coast is formed of different types of rock, the softer ones are destroyed first. This creates a coastline made up of bays divided by headlands. The headlands shelter the bays, allowing beaches to build up and protect the softer rock. Meanwhile, the shape of the coast concentrates the wave energy on the headlands, creating caves, arches, and stacks.



## SEA CAVES

Where hard, strong rock is being undercut by wave action, the process can create sea caves at the base of cliffs. Most of these sea caves are not very deep, because over time the waves crashing into them makes their ceilings collapse. But this process can also create dramatic blowholes, where breaking waves are forced up through gaps in the cave roof to form fountains of salty spray.